

Samsung KNOX™:

KNOX Glossary of Terms and Acronyms

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Introduction

The following are the common terms and acronyms used throughout the suite of Samsung KNOX™ documentation.

An Overview of Samsung KNOX

Samsung KNOX™ is the comprehensive enterprise mobile solution for work and play. With the increasing use of smartphones in businesses, Samsung KNOX addresses the mobile security needs of enterprise IT without invading the privacy of its employees.

Platform Security

Samsung KNOX addresses platform security with a comprehensive three-pronged strategy to secure the system: Customizable Secure Boot*, ARM® TrustZone®-based Integrity Measurement Architecture (TIMA), and a kernel with built-in Security Enhancements for Android (SE for Android) access controls.

Application Security

In addition to securing the platform, Samsung KNOX addresses enterprise application and data security requirements. Samsung KNOX container provides security for enterprise data by isolating enterprise applications and encrypting enterprise data both at-rest and in motion.

Mobile Device Management

Samsung KNOX works with enterprise-preferred MDM vendor solutions and provides industry-leading security and management controls

Glossary of KNOX Terms

Term	Definition
ABOOT	- The Application Bootloader (ABOOT) boots the Android kernel/Recovery kernel - Runs the ODIN protocol on the device. - Mechanism to download images onto the device from a host machine (like Windows/Linux PC).
Active Directory	- A directory service created by Microsoft for Windows domain networks. - A Microsoft Active Directory (AD) domain controller authenticates and authorizes all users and computers in a Windows domain type network. - Assigns and enforces security policies for all computers and installing or updating software.
AES-256 Encryption	- The AES (Advanced Encryption Standard) is used to securely encrypt uploaded files while they are temporarily hosted on a file server. - With AES-256, 256-bit encryption, currently the strongest available, is used.
APK	- An Android application package file (APK) is the file format used to distribute and install application software and middleware onto Google's Android operating system.
App Wrapping	- Task performed on an application to enable it to work inside a secure KNOX container. - Performed using Samsung's automated app wrapping service. - Binary-edited DEX and other modified files are combined back into an APK and signed. - The signing process for platform applications such as Contacts, Calendar, Email etc., differs slightly from third-party apps (e.g., Box, Salesforce):
ARM Trustzone®	- Provides continuous integrity monitoring of the Linux kernel via the TIMA security watcher. - Enables strong isolation to separate the code execution on a single physical processor core into two worlds, "secure world" and "normal world" (or "non-secure world").

Boot Loader/ Primary Boot Loader/ Secondary Boot Loader	- A computer program that loads the main operating system or runtime environment for the computer after completion of the self-tests. - The Primary Boot Loader (PBL) is permanently placed in a protected boot sector and executes directly after reset. - The PBL acts as a communication engine to download the Secondary Boot Loader (SBL) into the internal RAM and then activates it. - o The SBL adds functions for erase and programming of flash memory and EEPROM; it also handles the actual download of new or updated software. This allows a minimum ROM utilization by the PBL.
CAC	- A plastic Common Access Card (CAC) containing an integrated circuit card (ICC) or memory that securely stores personal identification information. - The digitally stored information is read by a smart card reader. - A contact smart card is inserted into the reader while a contactless smart card is brought within proximity of the reader. - The reader communicates with a software processing system that processes the data and completes the requested transaction.
Centrify	- Centrify provides multi-application SSO for mobile apps inside the Samsung KNOX container. - The KNOX SSO implementation is based on Centrify's Mobile SSO solution, which interacts with their Cloud service. - The multi-tenanted Centrify Cloud Service connects to each Enterprise's AD infrastructure using a plug-in. - SSO service is an APK provided by Centrify
CSB	- Customizable Secure Boot (CSB) allows the root-of-trust to be a government issued or approved certificate, rather than the default Samsung certificate. - This root-of-trust enables deployments in government installations.
DAC	- With Discretionary Access Control (DAC), the owner of the object specifies which subjects can access the object. - Control of access is based on the discretion of the owner. - o When creating data, the owner decides what access privileges to give to other users when they attempt to access the data. - o The operating system will then make the access control decision based on the access privileges created. - o Contrast to Mandatory Access Control (MAC),
DAR	- Data-at-Rest (DAR) refers to inactive data which is stored physically in any digital form (e.g. servers, hard drives, mobile devices, etc.).
Data Attack	- A special type of attack that does not require the modifying or loading of code. - Relies on data vulnerability.
Denial Log	- Captures SE for Android denials in Enforcement mode on the device and uploads them to a Samsung Server. - These denial logs are stored on the device at: <i>/data/misc/audit/audit.log</i> and <i>audit.old</i>.

DIT	- Data-in-Transit (DIT) refers to data that is being transmitted over a network (e.g. email, file transfers, account log in, etc.). - Biggest threats are interception and alteration. - Also known as Data-in-Motion (DIM)
EDM	- The Enterprise Device Manager (EDM) Framework provides an 'Enterprise SSO Policy' that can be used by MDM clients to provision the SSO service. - The EDM in-turn passes the configuration and provision information to the SSO service on the device via an 'Enterprise Security Manager'.
Enforcing Mode	- SE for Android is enforcing the loaded policy.
FIPS 140-2	- The Federal Information Processing Standard (FIPS) Publication 140-2 is a US security standard that helps ensure companies that collect, store, transfer, share and disseminate sensitive but unclassified information and controlled unclassified information. - Defines four levels of security, named "Level 1" (lowest) to "Level 4" (highest). - Samsung KNOX is FIPS 140-2 certified. - Samsung KNOX meets the requirements for FIPS 140-2 Level 1 certification for both data-at-rest (DAR) and data-in-transit (DIT).
FLE	- File Level Encryption (FLE) is a form of disk encryption where individual files or directories are encrypted by the file system itself. - o Contrast to full disk encryption where the entire partition or disk, in which the file system resides, is encrypted. - Uses FIPS-certified Kernel Crypto module. - Used to encrypt a KNOX container - Used to encrypt an external SD card
Kernel Compilation Flag	- The flag used to enclose the TIMA modifications to the kernel code.
IKEv1	- Internet Key Exchange 1 is the Main and aggressive IKE exchange modes with pre-shared key, certificates, Hybrid RSA, and EAP-MD5 authentications are supported.
IKEv2	- Internet Key Exchange 2 is configured with pre-shared key, certificates, EAP-MD5, and EAP-MSCHAPv2 authentication methods.
IPSec	- Internet Protocol Security (IPsec) VPNs grant or deny access to the corporate network as a whole, based on information at the network (routing/IP) layer. - Standard for Virtual Private Networks that uses the network cryptographic protocols for protecting IP traffic to provide an encrypted, secure "tunnel" for IP data traffic across a non-secure public extranet or the Internet. - Allows for trusted data to pass through networks which would otherwise be considered insecure.
ISA	- Integrity Service Agent (ISA) scans 3rd party APKs and uses Integrity Service Layer (ISL) services to scan platform and to store the baselines.

Isolation	○ Data isolation: Restricting access to content providers only from within the container. ○ Interaction isolation: Restricting containerized applications from interacting with applications outside the container. ○ Service isolation: Specified system services that are isolated to restrict functionality within the container.
LKM	- Loadable Kernel Modules (LKMs) are object files that contains code to extend the running kernel of an operating system.
MAC	- Mandatory Access Control (MAC) allows users to elevate their permissions to run certain commands as if they were the root user of the system. - MAC permissions are centrally managed by a central admin and not by the user.
MCM	- The Mobile Container Manager (MCM) is a part of the Samsung Enterprise Mobile Device Management (MDM) SDK add-on, an application development framework that enables an enterprise to create an MDM agent that Enterprise IT departments use to manage Samsung KNOX-enabled mobile devices. - MCM enables centralized management capabilities supporting secured containers to provide users with single sign-on for seamless access to mobile and web applications within the container.
MDM	- The Mobile Device Manager (MDM) provides expanded device management support for both Android and iOS platforms with full control over devices, including auto-management of ActiveSync mailbox access control lists.
Mocana's NanoSec™	- Standards-based full featured and RFC-compliant IPsec toolkit.
Normal World	- Features a system environment that encompasses all non-secure devices and software. - The normal world is intended for other regular operations.
NSA Suite B Cryptography	- A set of cryptographic algorithms announced by the National Security Agency (NSA) as part of its Cryptographic Modernization Program. - Serves as an interoperable cryptographic base for both unclassified information and most classified information. - One component is Advanced Encryption Standard (AES), with key sizes of 128 and 256 bits.
ODE	- On Device Encryption (ODE) provides the ability to encrypt data residing on users' devices so that it can't be read by anyone other than the authorized user. - ODE protects any local data such as customer information, confidential corporate information and contacts located in a device's internal memory, or stored on an external SD card.
ODIN	- A program run on a PC that can load and flash firmware image files ("ROMs") onto Samsung smartphones. - Can communicate with a smartphone using USB.
Permissive Mode	- SE for Android has loaded the policy, but is not enforcing it: - Generally used for testing as the audit log will contain the AVC-denied messages ○ The audit log can be used to determine the cause and possible resolution by generating appropriate allow rules.
PKCS	- Public Key Cryptography Standards (PKCS) is a group of standards devised and

	published by RSA Security Inc. with the intent of making secure information exchange on the Internet possible using a public key infrastructure (PKI). - KNOX platform provides applications access to the hardware certificates on the CAC via standards-based PKCS APIs.
PKI	- Public Key Infrastructure (PKI) certificates mandated by the US Department of Defense (DoD) for employees to “sign” documents digitally, encrypt and decrypt email messages, and establish secure online network connections. - Samsung KNOX allows the PKI certificates to be stored securely on the mobile device (software certificates) or be retrieved from a CAC (hardware certificates).
Policy Formulations	- SE for Android policies are crucial to guaranteeing that damage caused by any compromised application or service is contained. - Works in conjunction with MAC to ensure that all legitimate apps run properly, while allowing only “just enough” permissions. - Policy needs to evolve as the system evolves and features are added.
Primary Boot Loader	(See Boot Loader)
Proxying	- The required proxies intercept calls to the framework and other applications to include KNOX logic before forwarding the call to the original destination. - Consists of proxy classes that are replicas of Android components but provide additional or alternate behaviors to support containerization - The Wedge Framework is an extension of the Android framework that implements essential functionality for containerization. - o Also performs “reverse-proxying”
SaaS	- Software as a Service (SaaS) is a software distribution model in which applications are hosted by a vendor or service provider and made available to customers over a network, typically the Internet.
SAML	- SAML (Security Assertion Markup Language) is an XML standard that allows secure web domains to exchange user authentication and authorization data. Using SAML, an online service provider can contact a separate online identity provider to authenticate users who are trying to access secure content. - Used for Software as a Service (SaaS) applications such as Salesforce.com, Box, etc.

SE for Android	- Security Enhancements for Android (SE for Android) is a port of SE Linux to Android. - SE for Android provides an enhanced mechanism to enforce the separation of information based on confidentiality and integrity requirements. - o Incorporates a strong, flexible Mandatory Access Control (MAC) architecture into the major kernel subsystems and isolates applications and data into different domains. - This architecture prevents a compromise in one domain from propagating to other domains or the underlying mobile operating system (OS). - o Reduces threats of tampering and bypassing of application security mechanisms. - o Minimizes the amount of damage that can be caused by malicious or flawed applications. - SE for Android includes a set of security policy configuration files designed to meet common, general-purpose security goals.
Secondary Boot Loader	(See Boot Loader)
Secure World	- Contains all secure devices and software. - Intended for (infrequent) security sensitive operations.
Security Token	- Used to prove one's identity electronically. The token is used in addition to or in place of a password to prove that the customer is who they claim to be. The token acts like an electronic key to access something. - The Centrify service on the device manages the security token and is stored in file system. - The token is usually short-lived (30 minutes by default). - The token also stores keys and certificates in a standard key store file.
SSO	- Single Sign-On (SSO) is an authentication process that permits a user to enter one name and password in order to access multiple applications. - The process authenticates the user for all the applications they have been given rights to and eliminates further prompts when they switch applications during a particular session. - The SSO service is implemented using Centrify's Mobile Authentication Service (MAS) solution. - KNOX supports SSO support for apps within the Container.
Split Tunneling Mode	- Allows a VPN user to access a public network (e.g., the Internet) and a local LAN or WAN at the same time, using the same physical network connection.
TIMA	- Samsung's TrustZone-based Integrity Measurement Architecture (TIMA) uses ARM TrustZone hardware and provides continuous integrity monitoring of the Linux kernel. - The ARM TrustZone hardware effectively partitions memory and CPU resources into a "secure" and "normal" world. - TIMA is used along with Secure Boot and SE for Android to form the first line of defense against malicious attacks on the kernel and core boot strap processes. - When TIMA detects that the integrity of the kernel is violated, it notifies the enterprise IT via MDM which can then take policy-driven action in response.
TIMA Command ID	- The command ID passed to the TZ side to execute the TIMA function.

TIMA-LKMAuth	- TIMA LKM authentication (TIMA-LKMAuth) only authorizes the kernel modules that will be loaded into the kernel.
TIMA-PKM	- TIMA Periodical Kernel Measurement (TIMA-PKM) detects changes to the base kernel code pages. - Periodically hashes certain kernel code pages and verifies if the hash values have changed from the default values.
Triple DES encryption	- Triple Data Encryption Algorithm (Triple DES) block cipher which applies the Data Encryption Standard (DES) cipher algorithm three times to each data block. - Provides a method of increasing the key size of DES to protect against such attacks, without the need to design a completely new block cipher algorithm.
TUN Interface	- A virtual interface used to capture packets for encryption.
X.509 certificates	- Defines what information can go into a certificate. - Binds a name to a public key value. - Associate a public key with the identity contained in the X.509 certificate. - Contains information about the certificate subject and the certificate issuer.

About Samsung Electronics Co., Ltd.

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